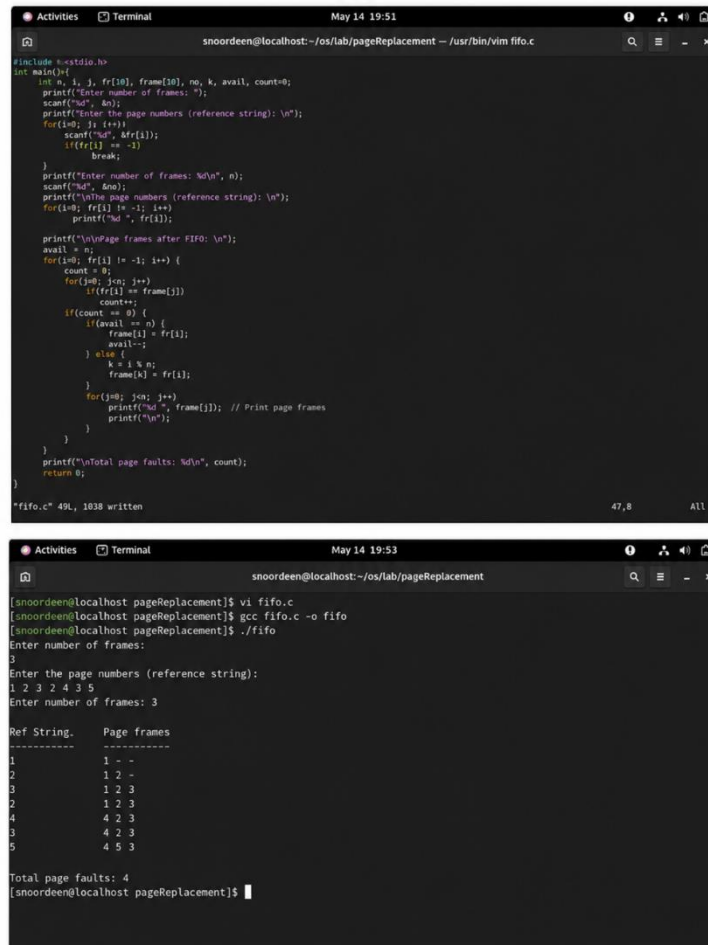


OPERATING SYSTEMS CONCEPT – LAB
EXP – 10: PAGE REPLACEMENT ALGORITHMS – FIFO & LRU

FIFO



```

snoordeen@localhost:~/os/lab/pageReplacement - /usr/bin/vim fifo.c
#include <stdio.h>
int main()
{
    int n, i, j, fr[10], frame[10], no, k, avail, count=0;
    printf("Enter number of frames: ");
    scanf("%d", &n);
    printf("Enter the page numbers (reference string): \n");
    for(i=0; i<n; i++)
    {
        scanf("%d", &fr[i]);
        if(fr[i] == -1)
            break;
    }
    printf("Enter number of frames: %d\n", n);
    scanf("%d", &n);
    printf("\nthe page numbers (reference string): \n");
    for(i=0; fr[i] != -1; i++)
        printf("%d ", fr[i]);
    printf("\n");

    printf("\nPage frames after FIFO: \n");
    avail = n;
    for(i=0; fr[i] != -1; i++) {
        count = 0;
        for(j=0; j<n; j++)
            if(fr[j] == frame[j])
                count++;
        if(count == 0) {
            if(avail == n) {
                frame[i] = fr[i];
                avail--;
            }
            else {
                k = i % n;
                frame[k] = fr[i];
            }
            for(j=0; j<n; j++)
                printf("%d ", frame[j]); // Print page frames
            printf("\n");
        }
    }
    printf("\nTotal page faults: %d\n", count);
    return 0;
}

*fifo.c" 49L, 1038 written
47,8 All

snoordeen@localhost:~/os/lab/pageReplacement
[snoordeen@localhost pageReplacement]$ vi fifo.c
[snoordeen@localhost pageReplacement]$ gcc fifo.c -o fifo
[snoordeen@localhost pageReplacement]$ ./fifo
Enter number of frames:
3
Enter the page numbers (reference string):
1 2 3 2 4 3 5
Enter number of frames: 3

Ref String.    Page frames
-----
1             1 - -
2             1 2 -
3             1 2 3
4             1 2 3
5             4 2 3
6             4 2 3
7             4 5 3

Total page faults: 4
[snoordeen@localhost pageReplacement]$
```